

MATH 55 SECOND MIDTERM EXAM, PROF. SRIVASTAVA
NOVEMBER 12, 2024, 5:10PM–6:30PM, 150 WHEELER.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: _____

Question	Points
1	8
2	6
3	6
4	8
5	9
6	6
7	7
Total:	50

Do not turn over this page until your instructor tells you to do so.
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Name and SID: _____

1. Select true or false (by filling in the circle) for each of the following. There is no need to provide an explanation.

(a) (2 points) If $n \geq 2$ and $1 \leq r \leq n - 1$, then $\binom{n}{r}$ is divisible by n .

☐ True

☐ False

(b) (2 points) If $n \geq 1$ and $0 \leq a \leq b \leq n$ then $\binom{n}{a} \leq \binom{n}{b}$.

☐ True

☐ False

(c) (2 points) If T is a tree then T has exactly two 2-colorings.

☐ True

☐ False

(d) (2 points) If $G = (V, E)$ is a simple graph then the number of distinct subgraphs of G is at least $2^{|E|}$.

☐ True

☐ False

[Scratch Space Below]

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2. (6 points) Prove by induction that for every integer $n \geq 4$,

$$2^n \leq n!.$$

[Scratch Space Below]

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3. (6 points) Consider the function $f : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$ recursively defined by $f(1) = 1$

$$f(n) = f(n-1) + 2n - 1, \quad n \geq 2.$$

Show that for all $n \in \mathbb{Z}_+$, $f(n)$ is not prime.

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4. (a) (4 points) Prove that if T is a tree with n vertices, then T must have exactly $n - 1$ edges. (You may not just cite that this was proven in class, but you may use other properties of trees which we proved).

- (b) (4 points) Prove that if $n \geq 1$ and $1 \leq k \leq n$, and G is an acyclic graph with n vertices and exactly k connected components, then G must have exactly $n - k$ edges. You may use the statement of part (a), even if you did not prove it.

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5. Recall that the degree sequence of a graph is a list of the degrees of its vertices in nondecreasing order. Give examples of graphs with the following properties, or prove that no such graph can exist.

(a) (3 points) A simple graph with degree sequence $2, 2, 1, 1$.

(b) (3 points) A tree with 5 vertices which has an Eulerian circuit.

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(c) (3 points) A simple graph with 4 vertices and 5 edges which is 2-colorable.

6. (6 points) Give a combinatorial proof that $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$.

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7. (a) (3 points) What is the cardinality of the set:

$$S = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \text{ are positive integers and } x_1 + x_2 + x_3 = 5\}?$$

Justify your answer.

- (b) (4 points) What is the cardinality of the set:

$$T = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \text{ are even positive integers and } x_1 + x_2 + x_3 \leq 11\}?$$

Justify your answer.

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[Scratch Paper]

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of 2-element subsets of V , called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y . The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \dots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \dots k$. The path is said to **connect** x_0, x_k , **visit** the vertices $x_0, \dots, x_k$, and **traverse** the edges $e_1, \dots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \rightarrow \{1, \dots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y . A graph is **k-colorable** if it has a k -coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G . A **connected component** of G is a maximal connected subgraph of G , i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the end-points. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.